

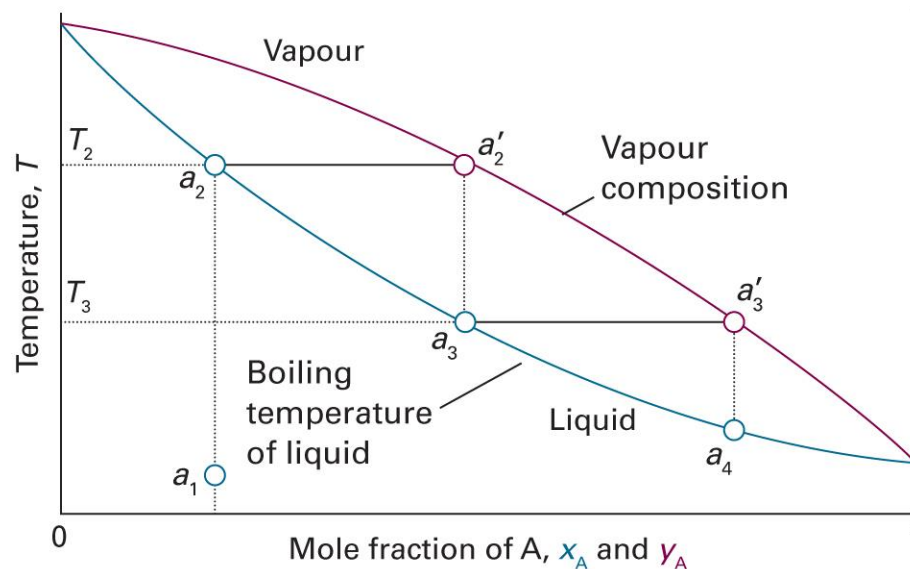
Physical Chemistry (I)

Lecture 18



Phase Diagrams of Real Solutions

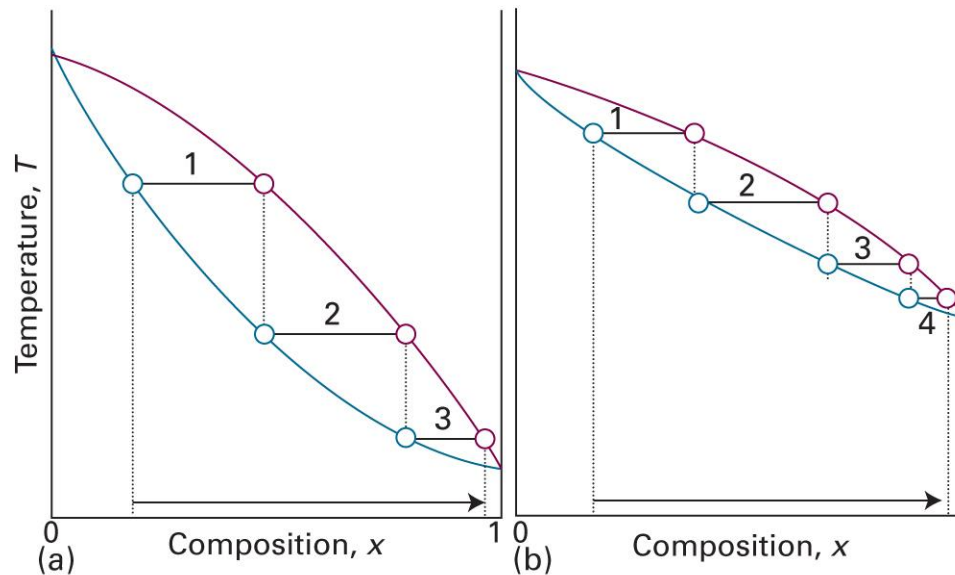
- **Liquid-liquid equilibrium:** phase separation can occur
- **Liquid-vapor equilibrium:** deviations from ideal behavior



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.4)

Distillation

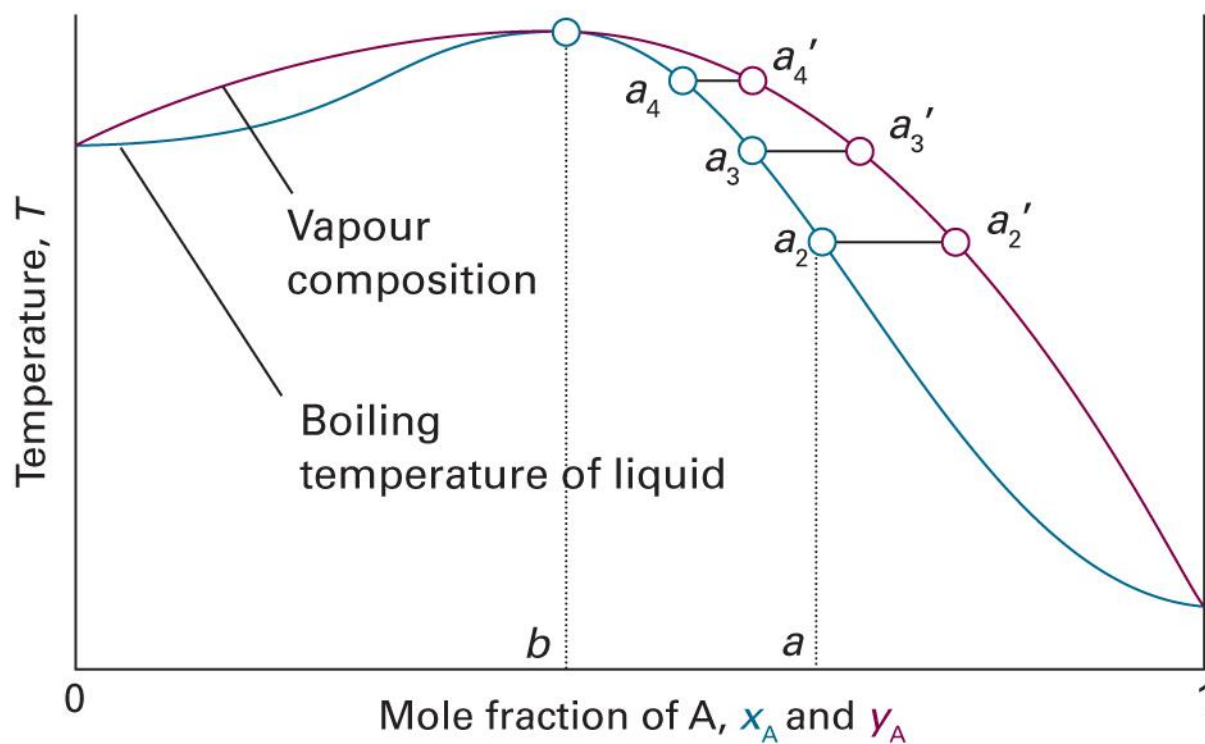
- **Simple distillation:** vapor withdrawn and condensed
- **Fractional distillation:** the cycle repeated successively



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.9)

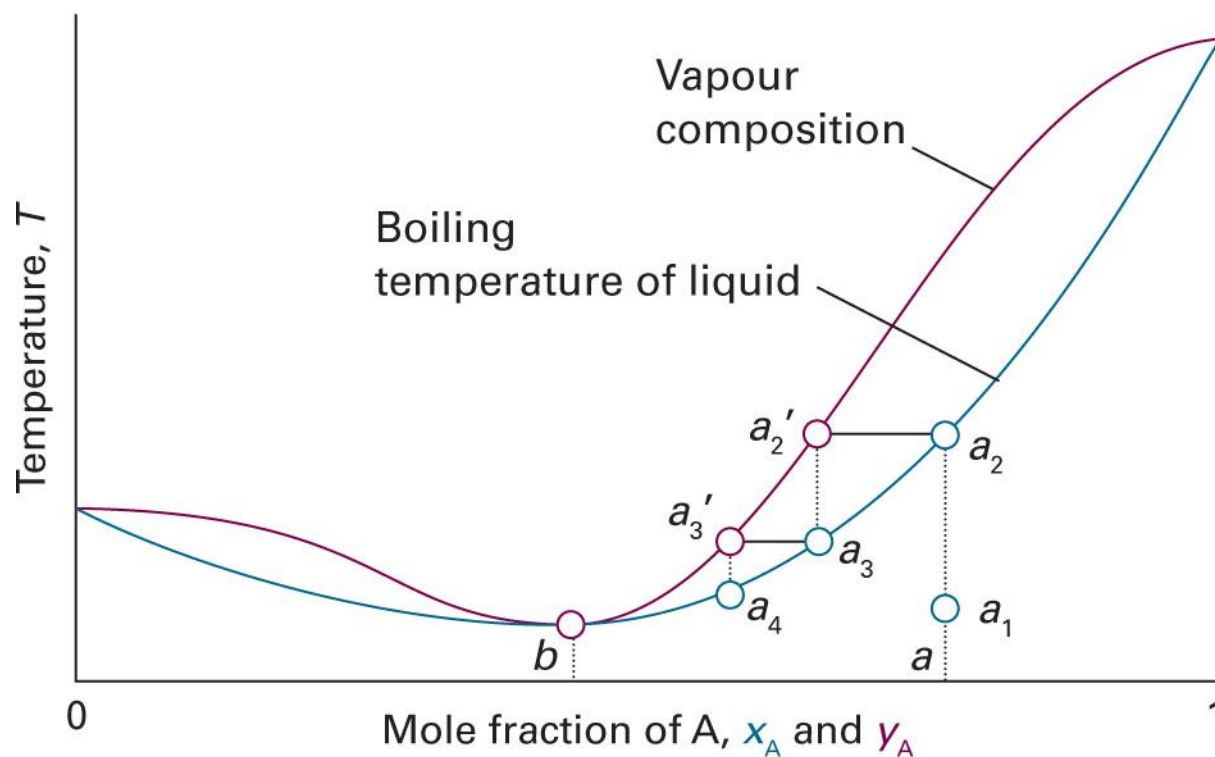
High-Boiling Azeotrope

$G^E < 0$ (A-B interactions stronger)



Low-Boiling Azeotrope

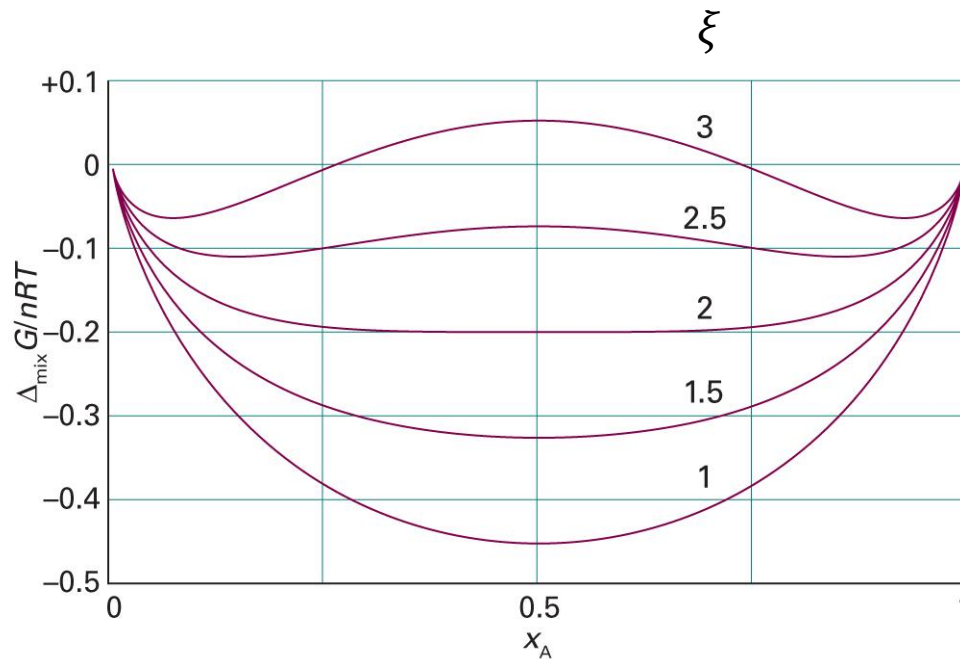
$G^E > 0$ (A-B interactions weaker)



(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 5C.11)

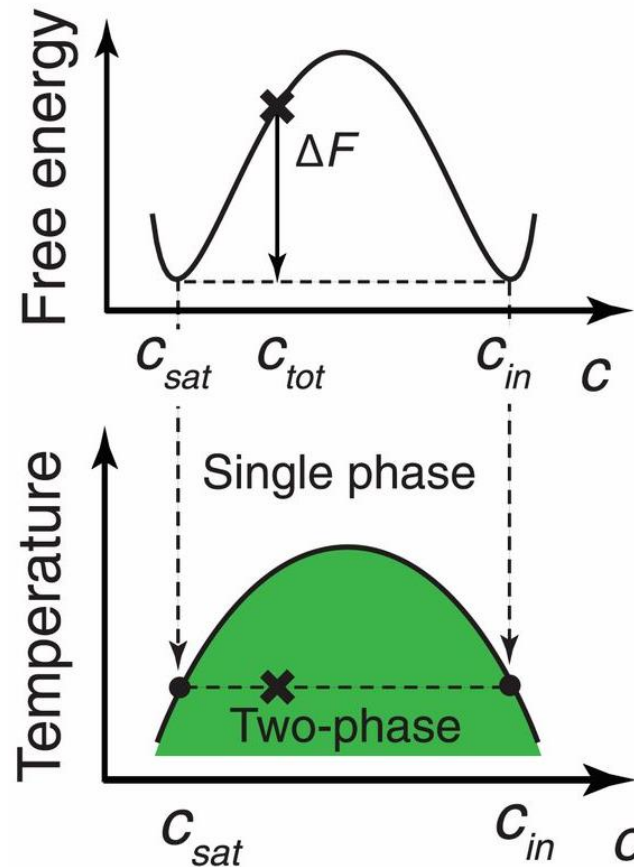
BW Model: Gibbs Energy of Mixing

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5B.5)

What Happens at $\xi > 2$?



(Image: Shin and Brangwynne, *Science* 357: eaaf4382 (2017), Figure 3)

Locations of Minima

$$\begin{aligned}\Delta_{\text{mix}}G &= nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B) \\ &= nRT\{x_A \ln x_A + (1 - x_A) \ln(1 - x_A) + \xi x_A(1 - x_A)\}\end{aligned}$$

The local minima can be found by using the derivative:

$$\begin{aligned}\left(\frac{\partial \Delta_{\text{mix}}G}{\partial x_A}\right)_{T,p} &= nRT\{\ln x_A + 1 - \ln(1 - x_A) - 1 + \xi(1 - 2x_A)\} \\ &= nRT\left\{\ln \frac{x_A}{1 - x_A} + \xi(1 - 2x_A)\right\} = 0\end{aligned}$$

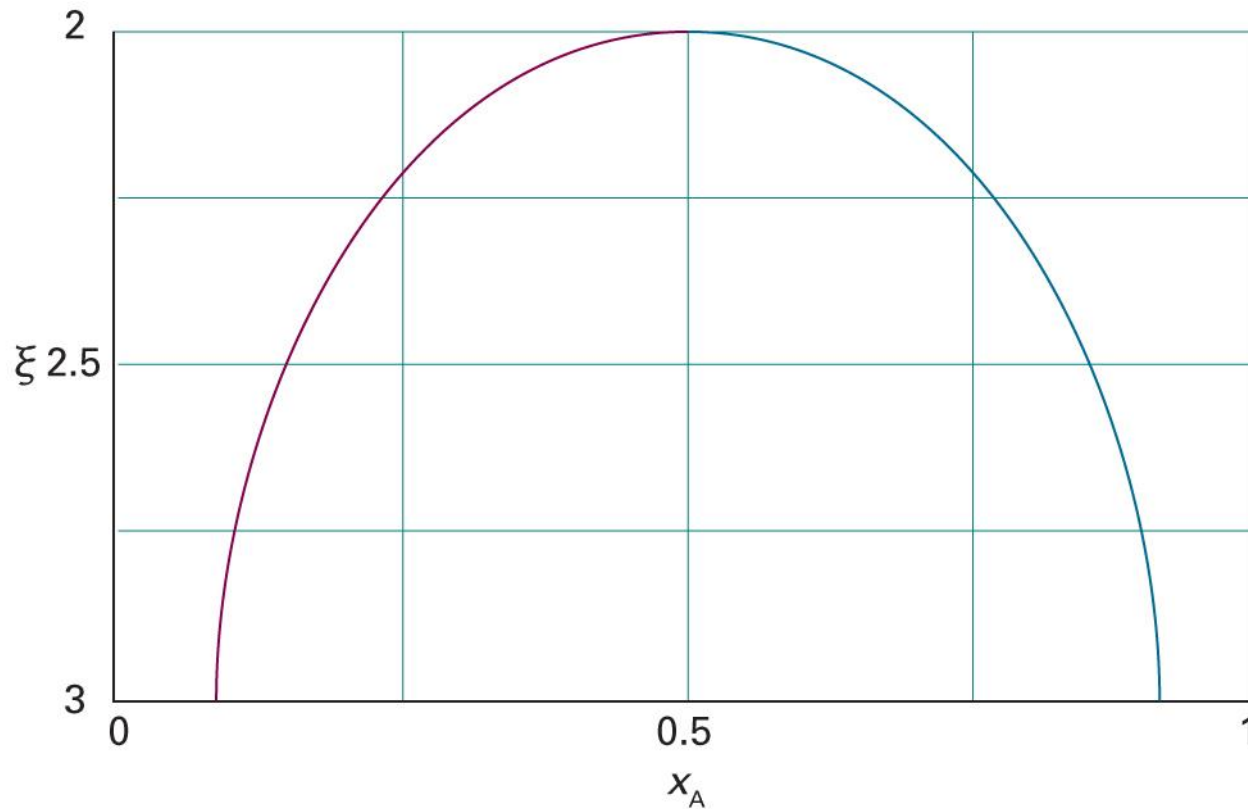
Locations of Minima

$$nRT \left\{ \ln \frac{x_A}{1 - x_A} + \xi(1 - 2x_A) \right\} = 0$$

The minima are located at x_A such that

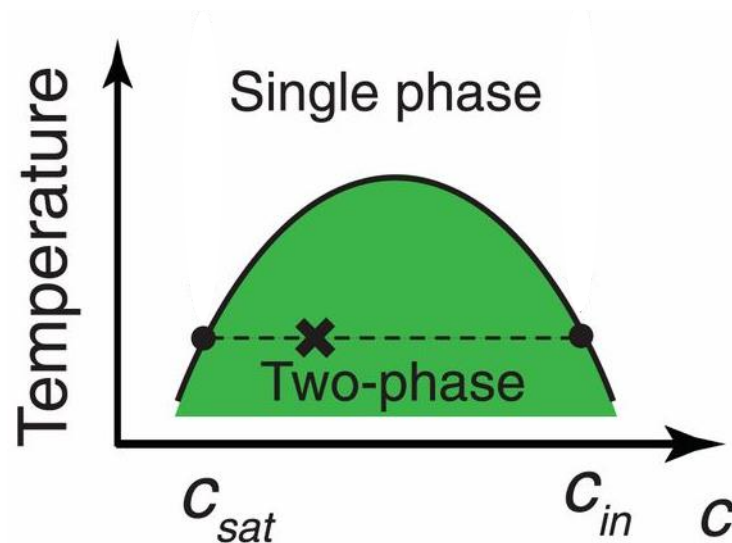
$$\ln \frac{x_A}{1 - x_A} = -\xi(1 - 2x_A)$$

Locations of Minima



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.17)

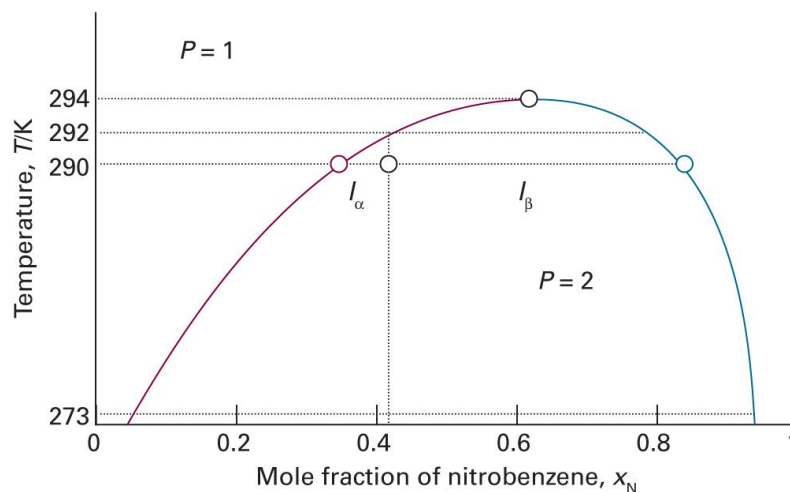
Phase Diagram



Lever rule applies in the two-phase region!

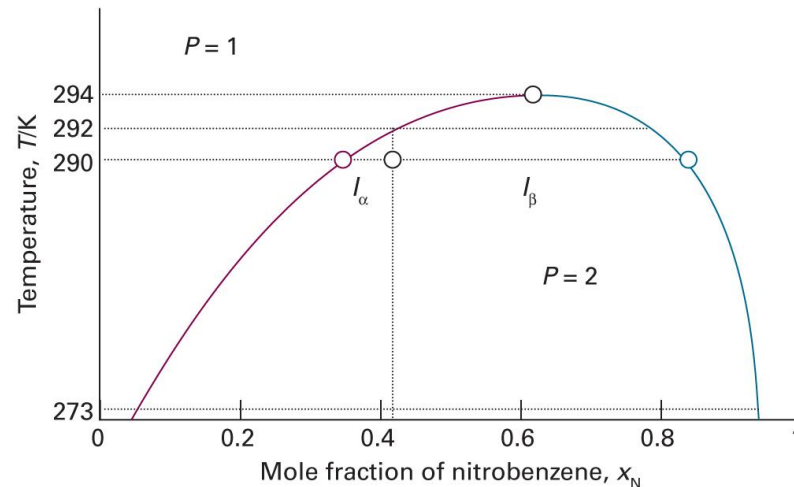
Example 5C.2

The phase diagram for the system nitrobenzene/hexane at 1 atm is shown below. A mixture of 50 g of hexane (0.59 mol C_6H_{14}) and 50 g of nitrobenzene (0.41 mol $\text{C}_6\text{H}_5\text{NO}_2$) was prepared at 290 K. What are the compositions of the phases, and in what proportions do they occur? To what temperature must the sample be heated in order to obtain a single phase?



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.14)

Example 5C.2



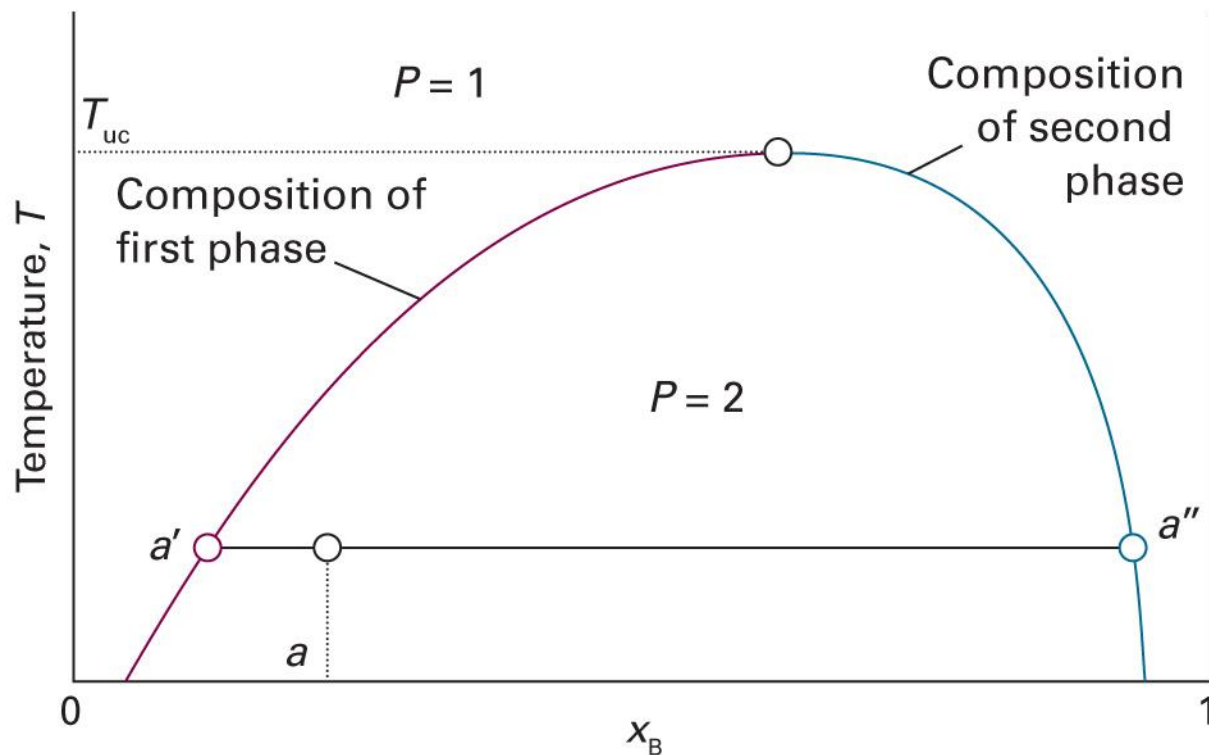
The tie line meets the phase boundaries at $x_N = 0.35$ and 0.83 .

Applying the lever rule, $n_\alpha : n_\beta = l_\beta : l_\alpha = (0.83 - 0.41) : (0.41 - 0.35) = 7:1$.

Thus, 7 times more hexane-rich phase than nitrobenzene-rich phase.

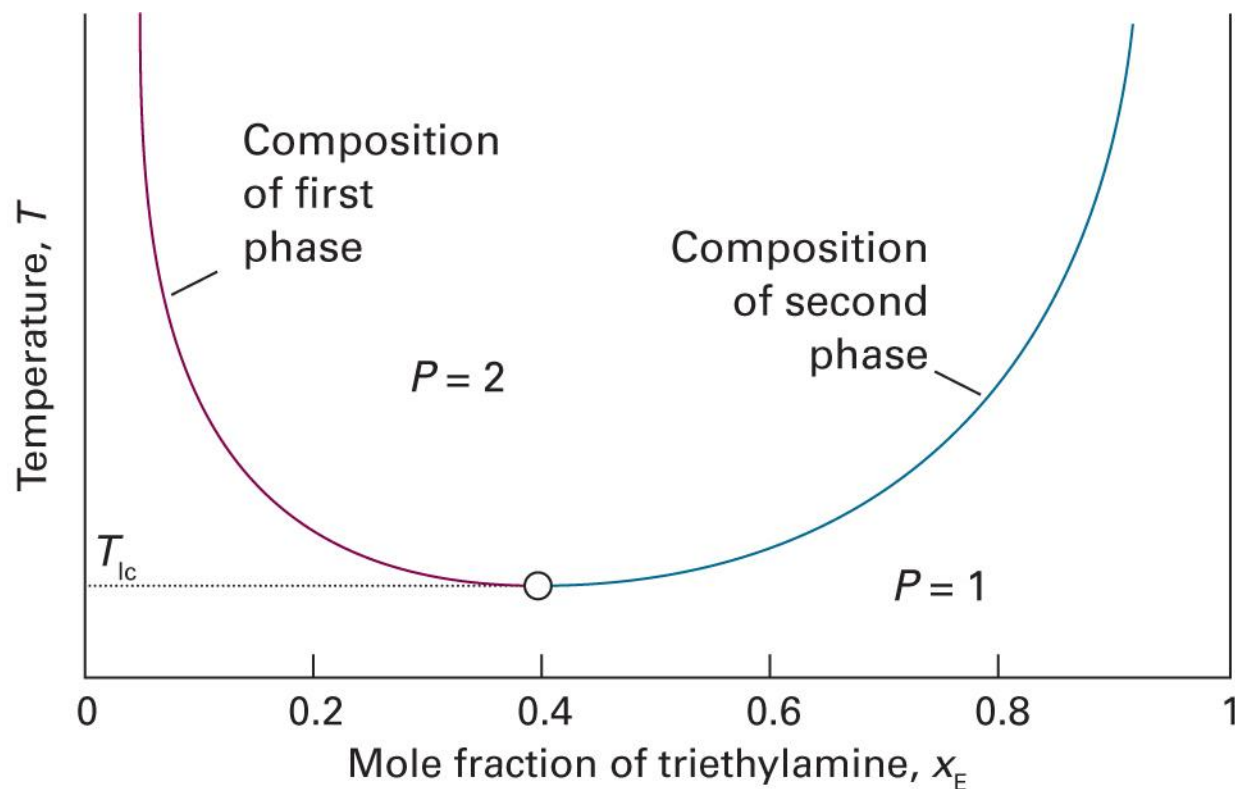
The isopleth enters into the single-phase region at $T = 292$ K.

Upper Critical Solution Temperature

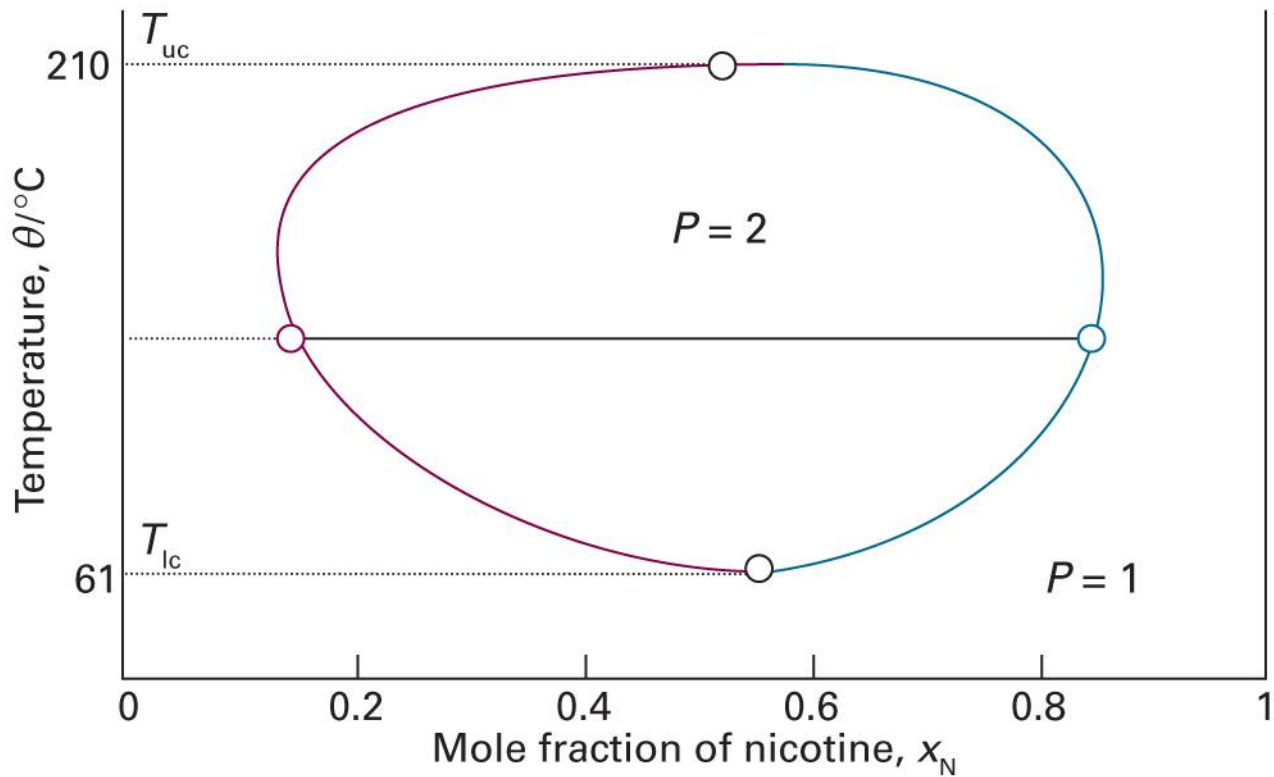


(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.13)

Lower Critical Solution Temperature



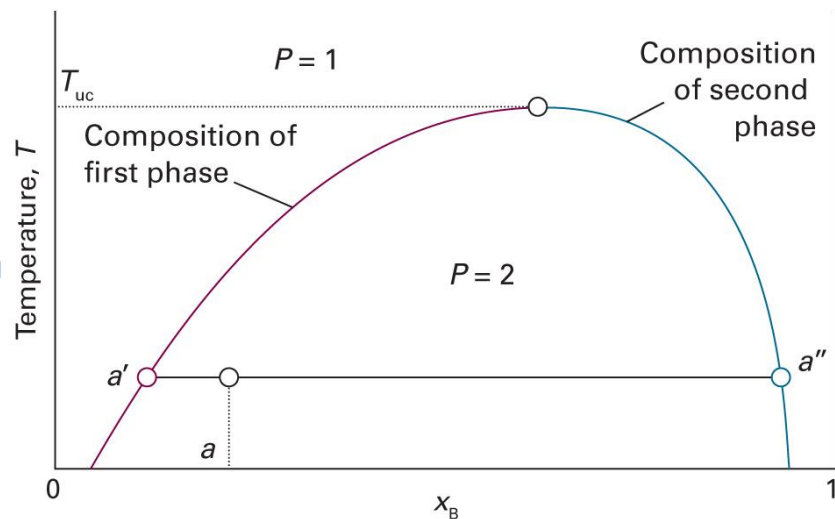
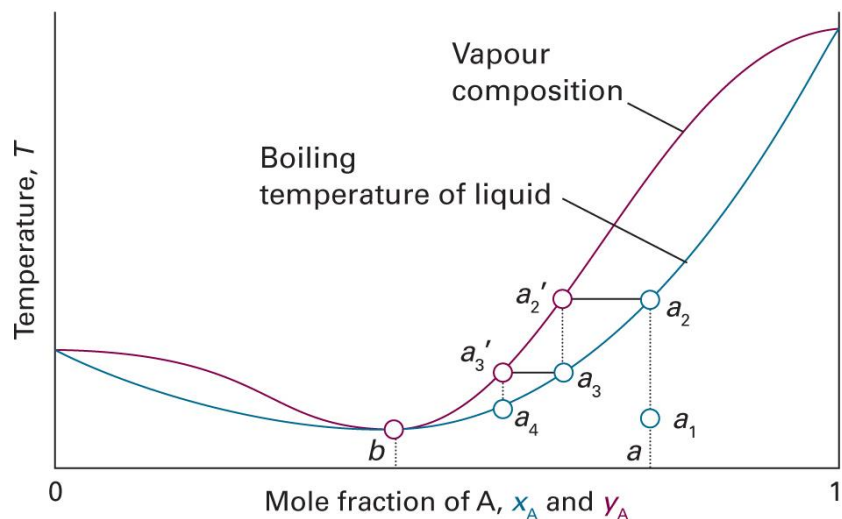
Both UCST and LCST



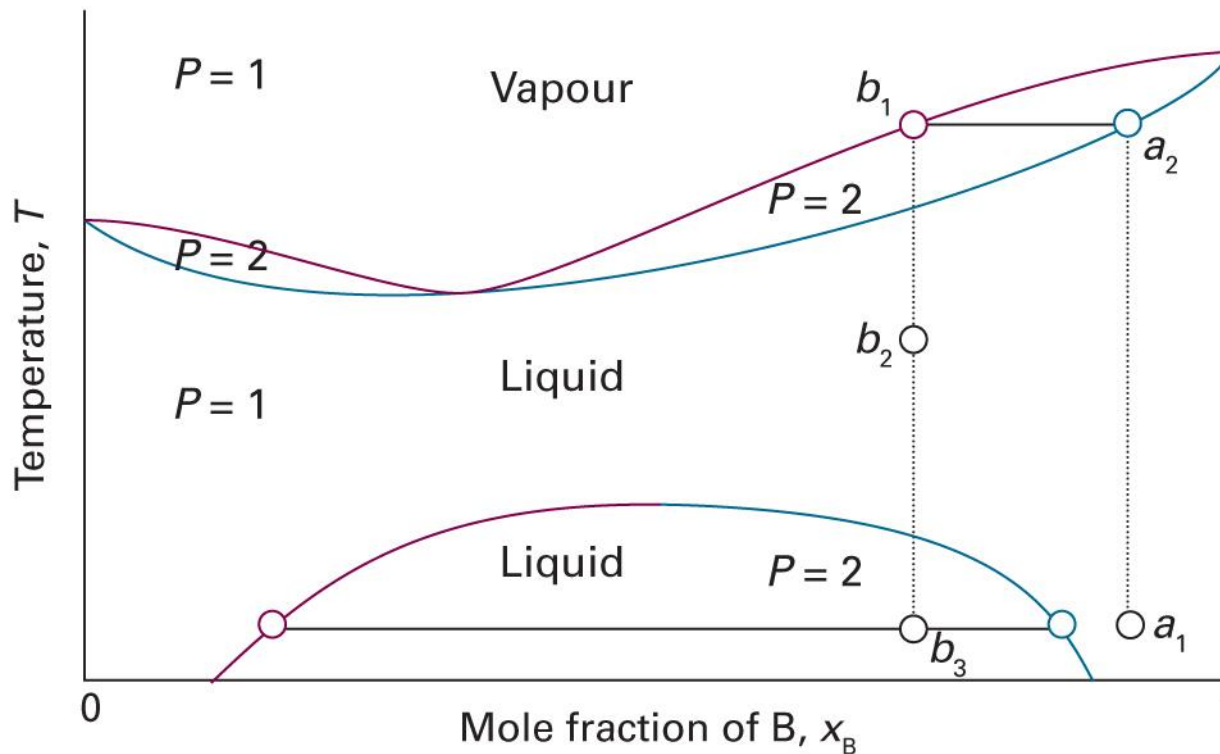
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.19)

L-G and L-L Equilibria

When A-B interactions are weaker than A-A and B-B

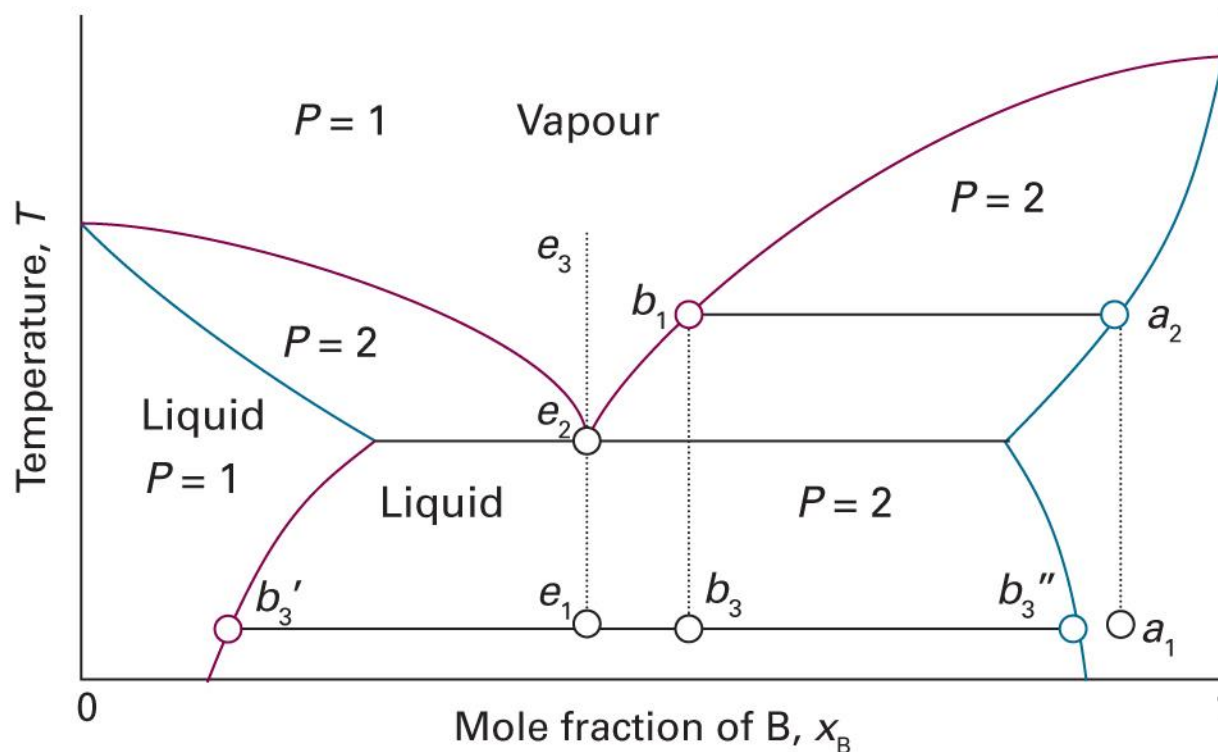


Case 1: Miscible before Boiling



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.20)

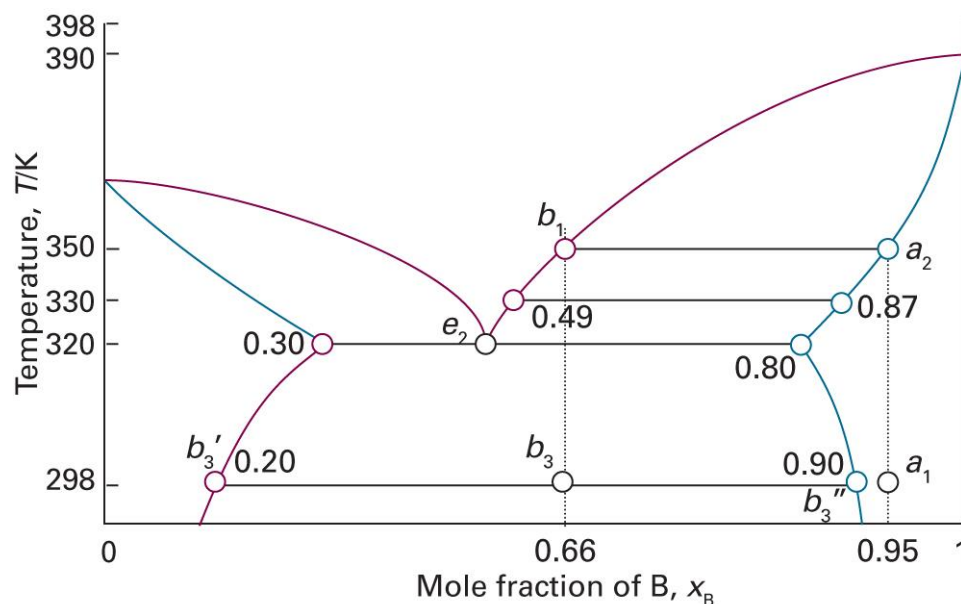
Case 2: Boiling before Miscible



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.21)

Example 5C.3

State the changes that occur when a mixture of composition $x_B = 0.95$ (a_1) in the figure below is boiled and the vapour condensed.



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.22)

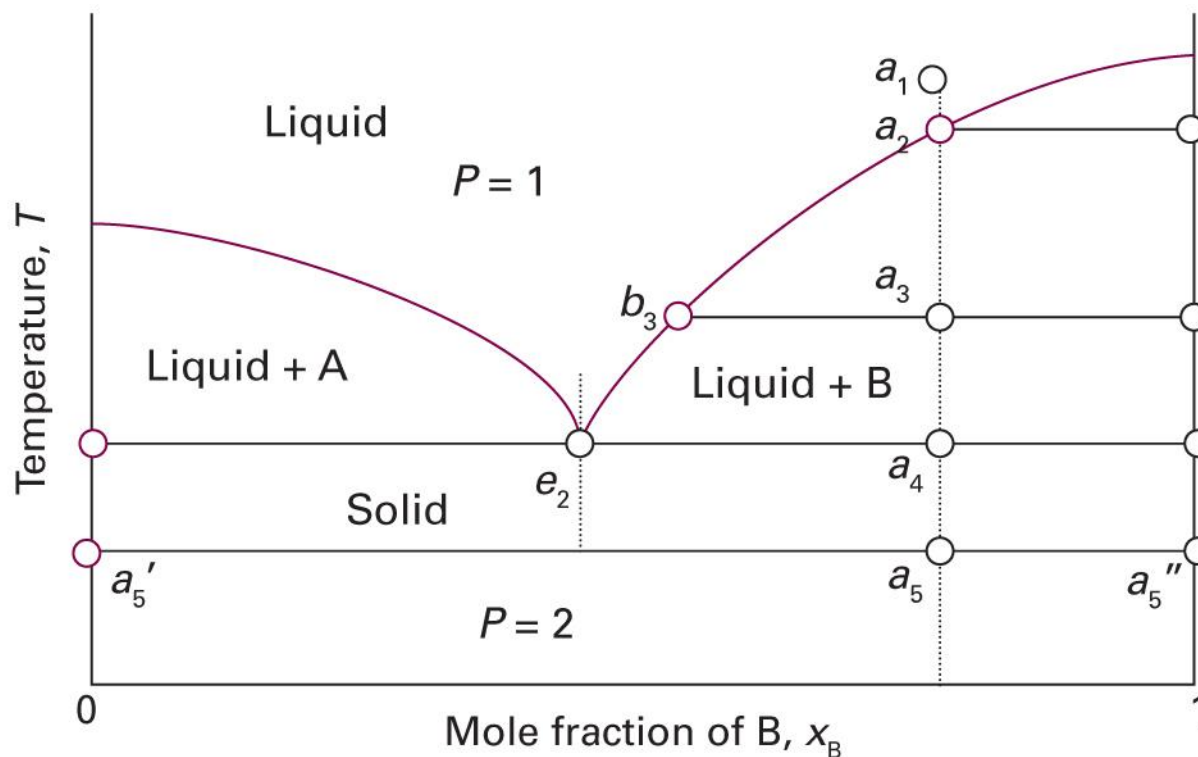
Liquid-Solid Equilibrium

Consider a two-component system (A and B)
where the two solids are almost immiscible.

Upon mixing,

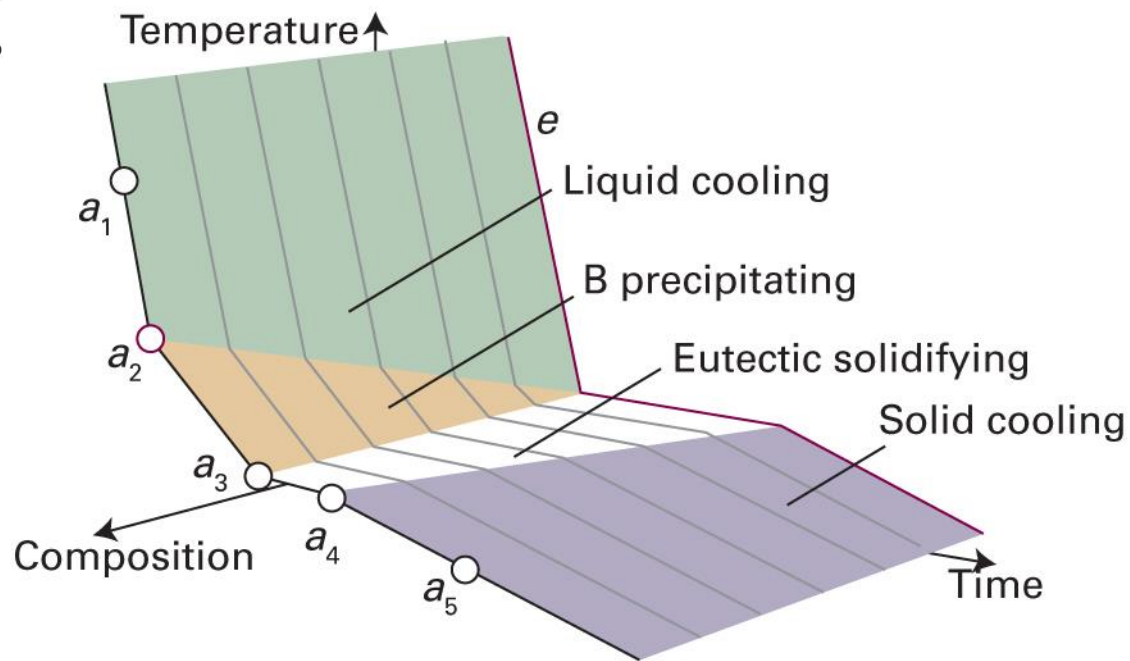
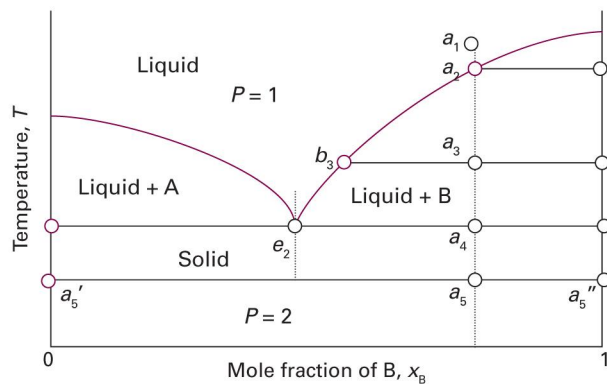
- Melting temperature ↓: A-B interactions are weaker
- Melting temperature ↑: A-B interactions are stronger

Phase Diagrams Involving Solids



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5D.1)

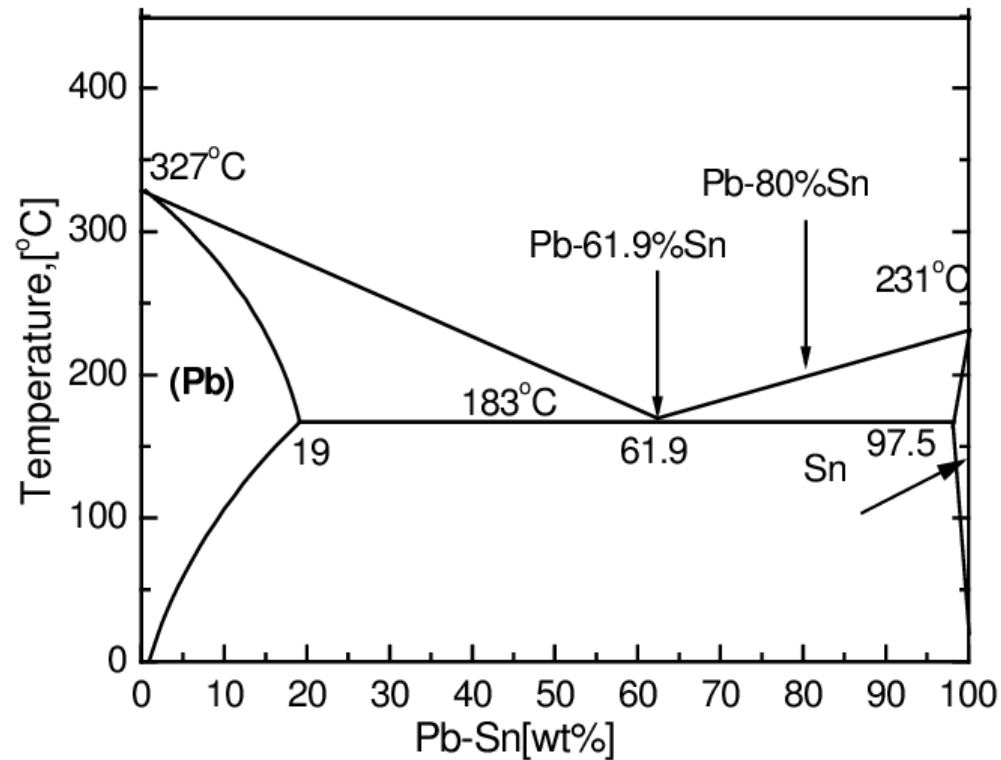
Eutectics



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 5D.1 and 5D.2)

Not in Atkins

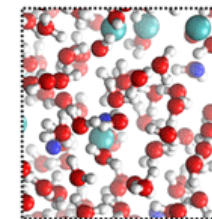
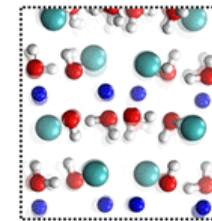
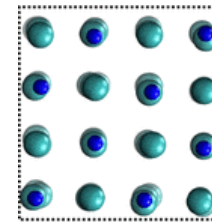
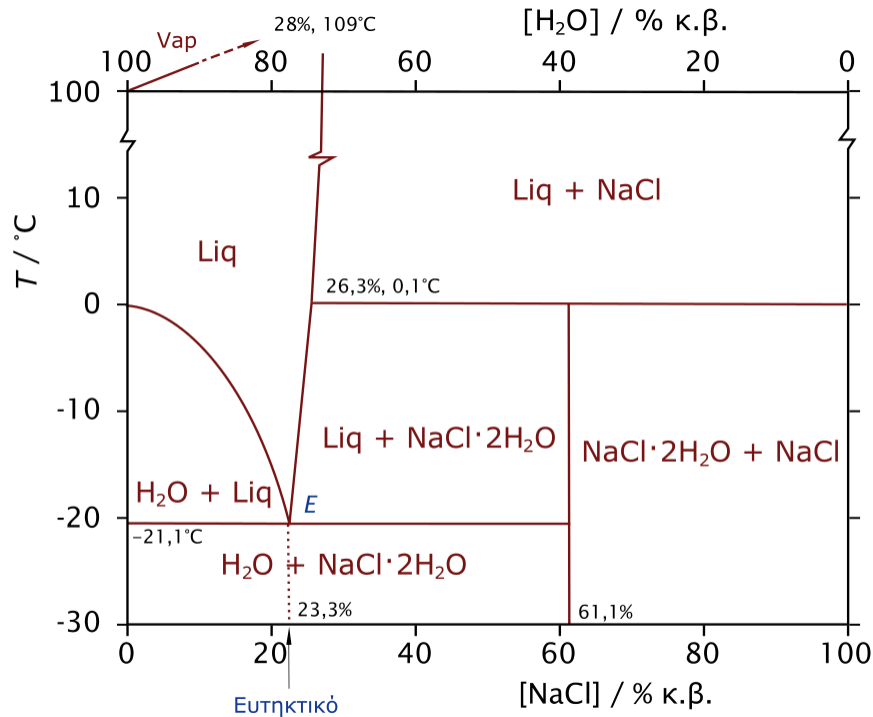
Pb-Sn System



(Image: https://www.researchgate.net/figure/The-Pb-Sn-phase-diagram_fig1_291162603)

Not in Atkins

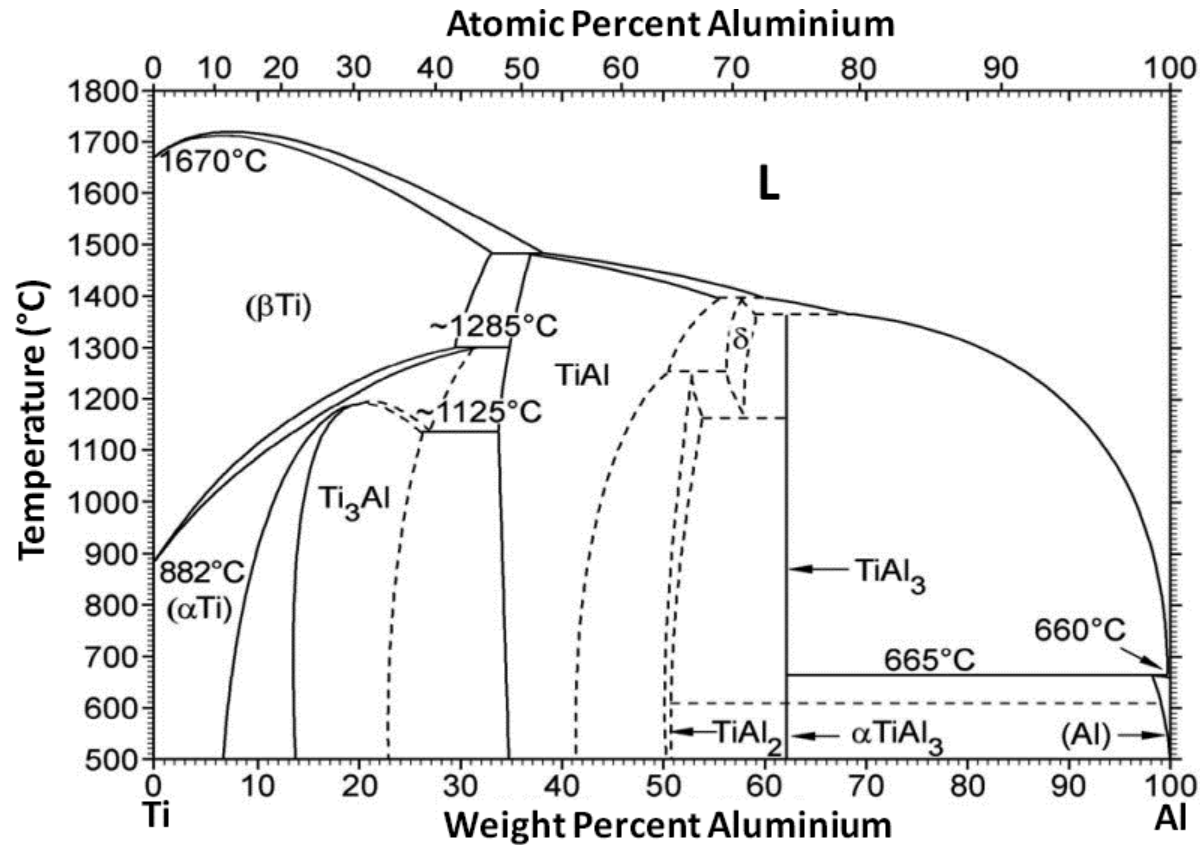
H₂O-NaCl System



(Image: <https://commons.wikimedia.org/wiki/File:H2O-NaCl-phase-diagram-greek.svg>)

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Ti-Al System



(Image: Mitra and Wanhill, *Structural Intermetallics* (2016), Figure 10)

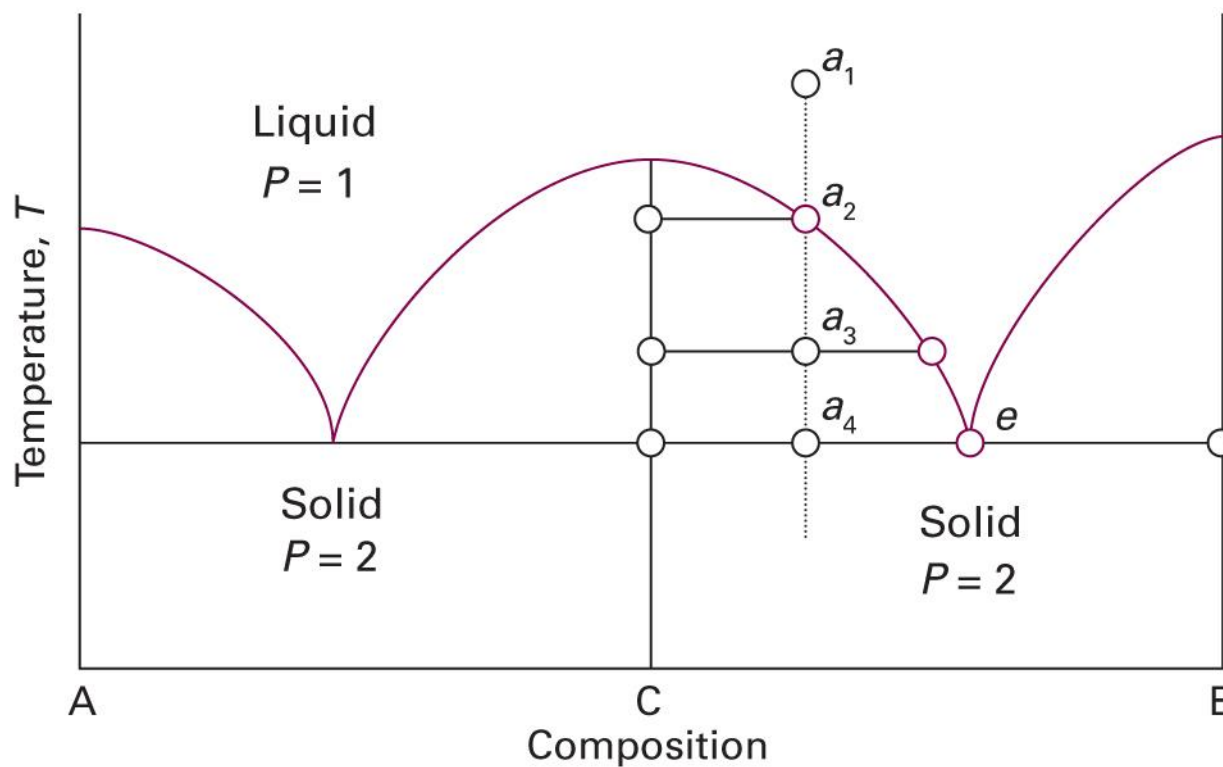
Reacting Systems

For reaction $A + B \rightarrow C$,

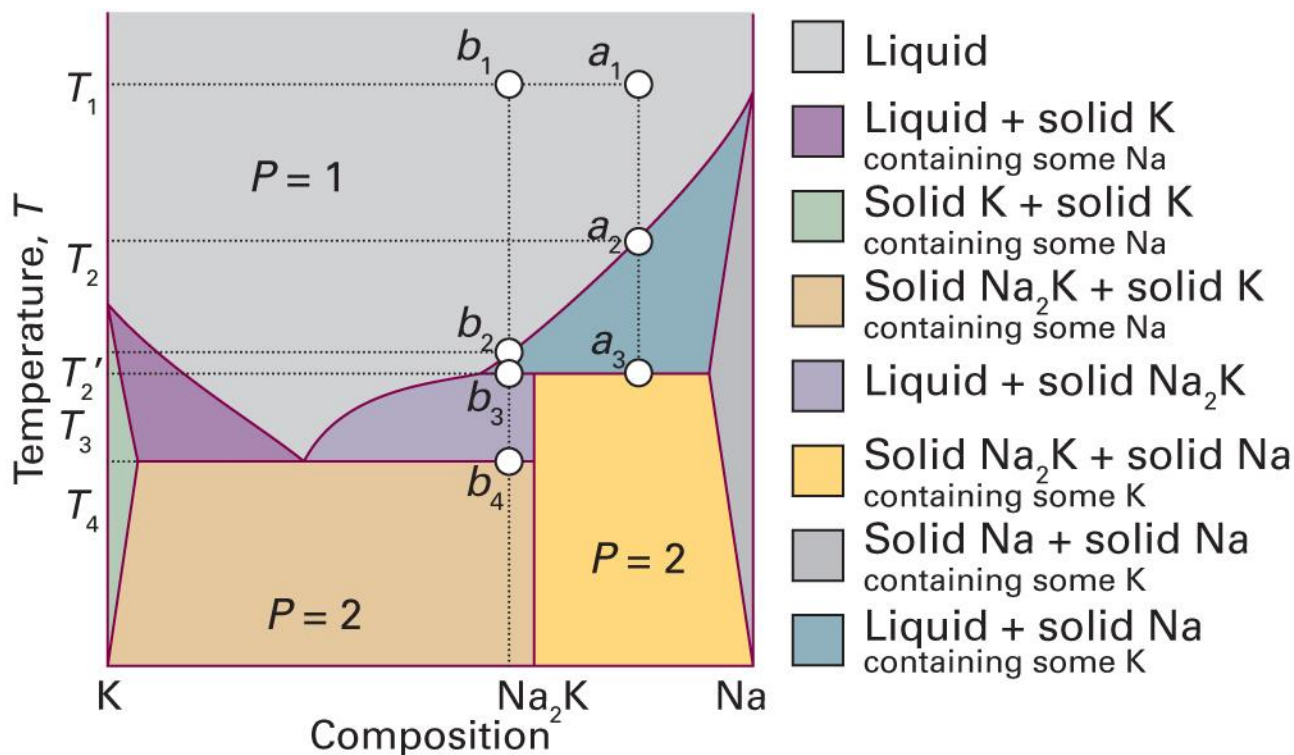
- A 100%, B 0% = A only
- A 75%, B 25% = A $\frac{2}{3}$, C $\frac{1}{3}$
- A 50%, B 50% = C only
- A 25%, B 75% = B $\frac{2}{3}$, C $\frac{1}{3}$
- A 0%, B 100% = B only

The x-axis can be (A $\rightarrow$ C $\rightarrow$ B)

Congruent Melting



Incongruent Melting



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5D.5)

Liquid-Solid Solutions

- Considered as ideal-dilute solutions
 - Liquid: solvent
 - Solid: solute
- Two assumptions
 - The solute is not volatile
 - The solute does not dissolve in the solid solvent

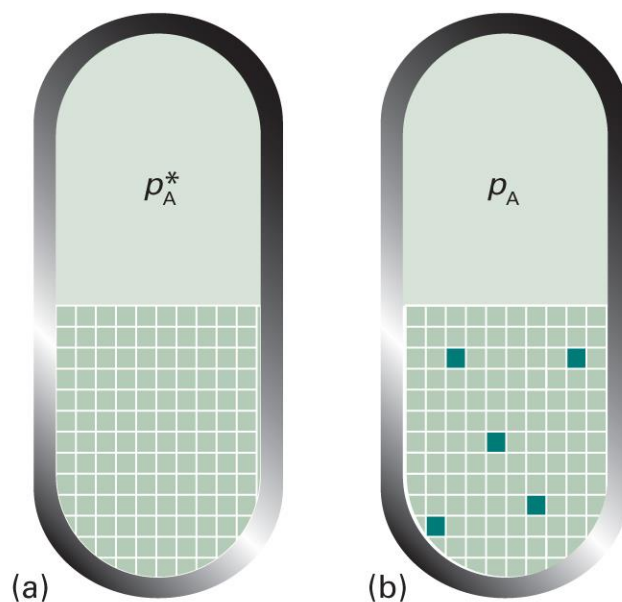
Colligative Properties

- Physical properties that depends on the relative number of solute particles present but not their chemical identity
1. Lowering of vapor pressure
 2. Elevation of boiling point
 3. Depression of freezing point
 4. Osmotic pressure



Origin of Colligative Properties

- Due to the **reduction of the chemical potential** of the liquid solvent as a result of the presence of solute



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5B.7)

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Lowering of Vapor Pressure

From Raoult's law,

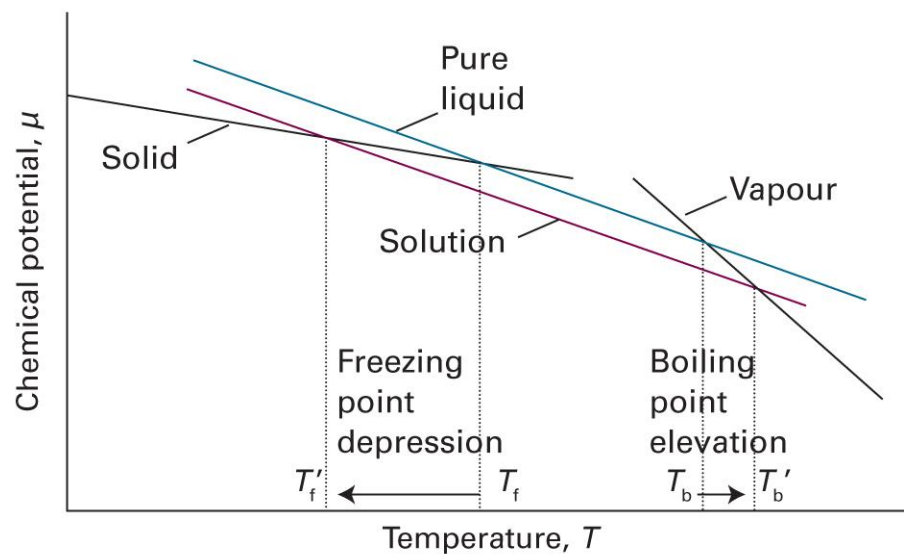
$$p_A = x_A p_A^*$$

The change in vapor pressure is

$$\Delta p_A = p_A^* - p_A = (1 - x_A) p_A^* = x_B p_A^*$$

Changes in Phase Transitions

$$\mu_A = \mu_A^* + RT \ln x_A \quad (x_A < 1)$$



Boiling: Liquid-Gas Equilibrium

$$\mu_A^*(g) = \mu_A(l)$$

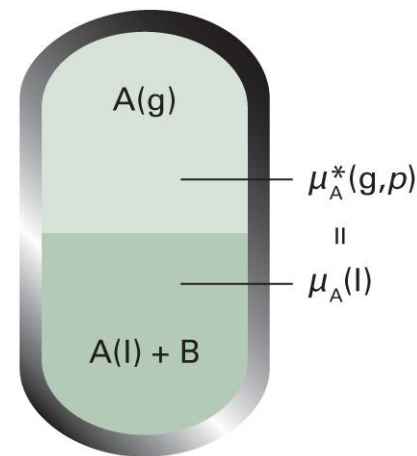
$$\mu_A^*(g) = \mu_A^*(l) + RT \ln x_A$$

which can be rearranged as

$$\ln x_A = \frac{\mu_A^*(g) - \mu_A^*(l)}{RT} = \frac{\Delta_{\text{vap}}G}{RT}$$

Using the Gibbs-Helmholtz equation,

$$\frac{d \ln x_A}{dT} = \frac{1}{R} \frac{d(\Delta_{\text{vap}}G/T)}{dT} = -\frac{\Delta_{\text{vap}}H}{RT^2}$$



Integration

$$\frac{d \ln x_A}{dT} = - \frac{\Delta_{\text{vap}} H}{RT^2}$$

$$d \ln x_A = - \frac{\Delta_{\text{vap}} H}{RT^2} dT$$

$$\int_{\text{pure solvent}}^{\text{current state}} d \ln x_A = - \frac{1}{R} \int_{\text{pure solvent}}^{\text{current state}} \frac{\Delta_{\text{vap}} H}{T^2} dT$$

$$\int_0^{\ln x_A} d \ln x'_A = - \frac{1}{R} \int_{T^*}^T \frac{\Delta_{\text{vap}} H}{(T')^2} dT'$$

Evaluation

$$\int_0^{\ln x_A} d \ln x'_A = -\frac{1}{R} \int_{T^*}^T \frac{\Delta_{\text{vap}} H}{(T')^2} dT'$$

Assuming that $\Delta_{\text{vap}} H$ is constant,

$$\ln x_A = -\frac{\Delta_{\text{vap}} H}{R} \int_{T^*}^T \frac{1}{(T')^2} dT' = -\frac{\Delta_{\text{vap}} H}{R} \left(-\frac{1}{T} + \frac{1}{T^*} \right)$$

$$\ln(1 - x_B) = \frac{\Delta_{\text{vap}} H}{R} \left(\frac{1}{T} - \frac{1}{T^*} \right)$$

Two (Valid) Approximations

$$\ln(1 - x_B) = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T} - \frac{1}{T^*} \right)$$

For $x_B \ll 1$, $\ln(1 - x_B) \approx -x_B$ and

$$x_B \approx \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T^*} - \frac{1}{T} \right) = \frac{\Delta_{\text{vap}}H}{RTT^*} (T - T^*)$$

Also, supposing $\Delta T_b = T - T^* \ll T^*$,

$$x_B \approx \frac{\Delta_{\text{vap}}H}{R(T^*)^2} \Delta T_b$$

Elevation of Boiling Point

$$x_B \approx \frac{\Delta_{\text{vap}}H}{R(T^*)^2} \Delta T_b$$

Rearrangement gives

$$\Delta T_b = K x_B$$

where $K = R(T^*)^2 / \Delta_{\text{vap}}H$.

Using the molality,

$$\Delta T_b = K_b b$$

where K_b is called the boiling-point constant.



Brief Illustration 5B.2

The boiling-point constant of water is $0.51 \text{ K kg mol}^{-1}$, and that of benzene is significantly larger, at $2.53 \text{ K kg mol}^{-1}$. For a solute present at a molality of 0.10 mol kg^{-1} , what would be the elevation of boiling point for each solvent?

$$\text{Water: } \Delta T_b = K_b b = (0.51 \text{ K kg mol}^{-1}) \times (0.10 \text{ mol kg}^{-1}) = 0.051 \text{ K}$$

$$\text{Benzene: } \Delta T_b = K_b b = (2.53 \text{ K kg mol}^{-1}) \times (0.10 \text{ mol kg}^{-1}) = 0.25 \text{ K}$$

Freezing: Solid-Liquid Equilibrium

$$\mu_A^*(s) = \mu_A(l)$$

$$\mu_A^*(s) = \mu_A^*(l) + RT \ln x_A$$

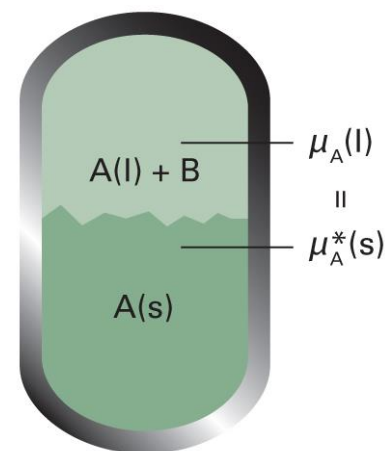
Similarly, we obtain

$$\Delta T_f = K' x_B$$

where $K' = R(T^*)^2 / \Delta_{\text{fus}}H$, or,

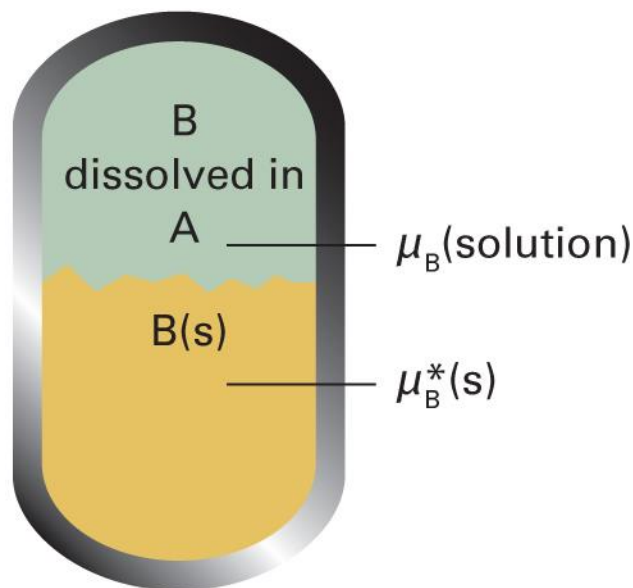
$$\Delta T_f = K_f b$$

where K_f is called the freezing-point constant.



Solubility

- **Saturation:** the undissolved solute in equilibrium with the dissolved solute



Chemical Potential of Solute

For solute B,

$$\mu_B^*(s) = \mu_B(l) = \mu_B^*(l) + RT \ln x_B$$

Rearrangement leads to

$$\ln x_B = \frac{\mu_B^*(s) - \mu_B^*(l)}{RT} = -\frac{\Delta_{\text{fus}}G}{RT}$$

Differentiating both sides and integrating,

$$\int_0^{\ln x_B} d \ln x'_B = \frac{1}{R} \int_{T_f}^T \frac{\Delta_{\text{fus}}H}{(T')^2} dT'$$

Solubility

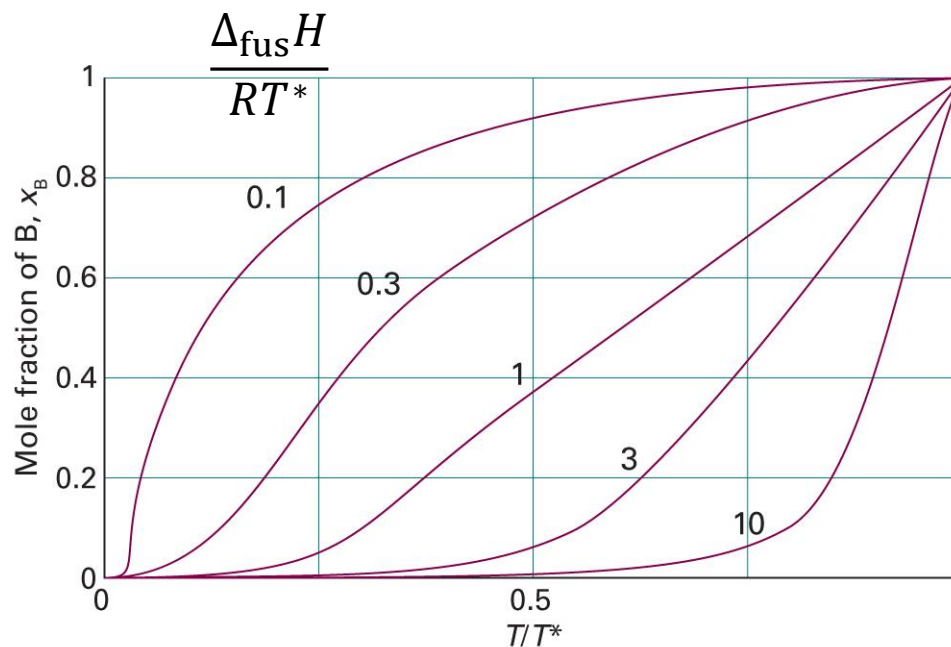
$$\int_0^{\ln x_B} d \ln x'_B = \frac{1}{R} \int_{T_f}^T \frac{\Delta_{\text{fus}} H}{(T')^2} dT'$$

Assuming that $\Delta_{\text{fus}} H$ is constant,

$$\ln x_B = \frac{\Delta_{\text{fus}} H}{R} \left(\frac{1}{T_f} - \frac{1}{T} \right)$$

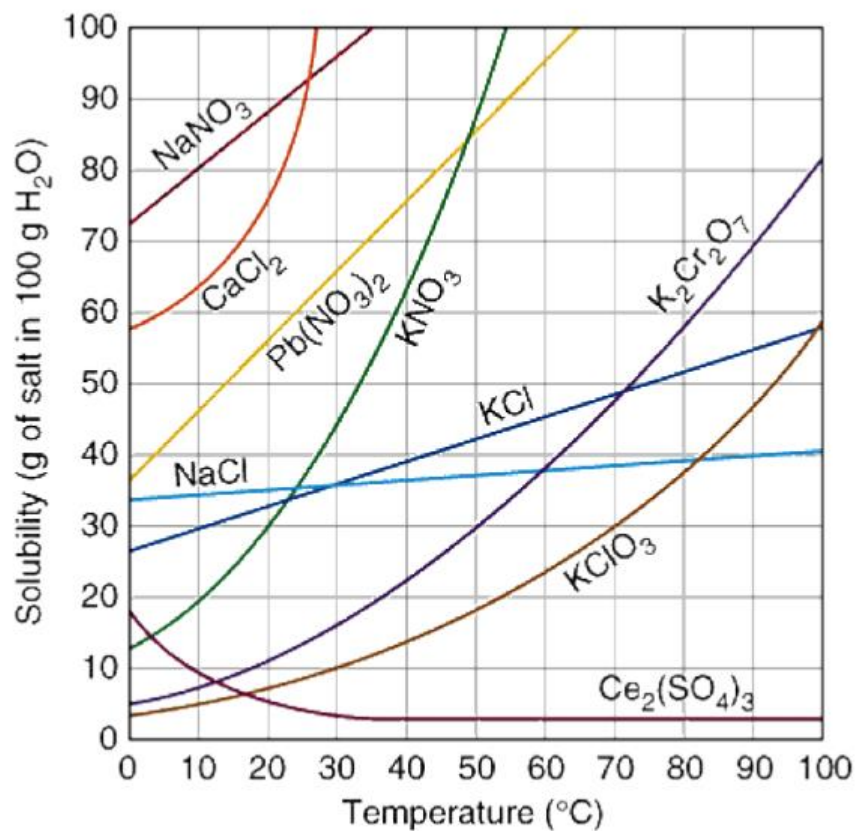
(Idealized) Solubility

$$\ln x_B = \frac{\Delta_{\text{fus}}H}{R} \left(\frac{1}{T^*} - \frac{1}{T} \right)$$



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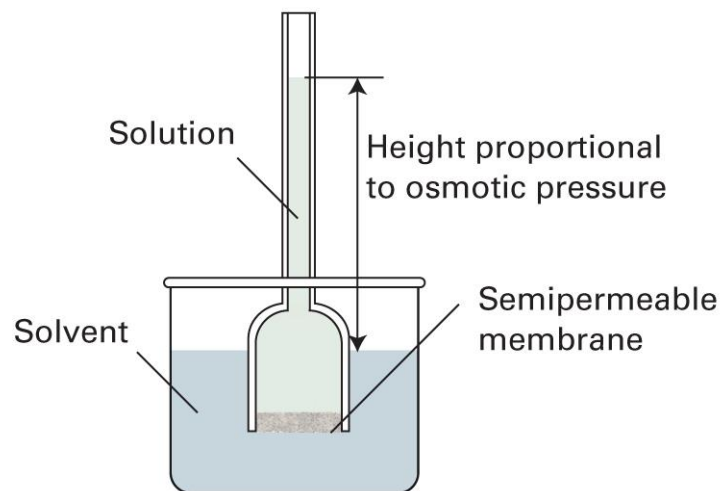
Real Solubility Data



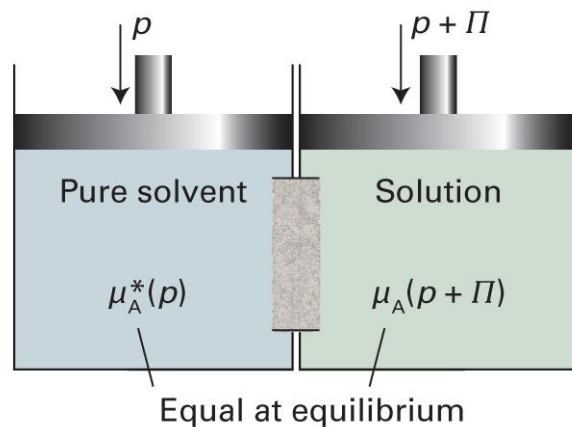
(Image: <https://socratic.org/questions/what-are-solubility-curves>)

Osmosis

the spontaneous passage of a pure solvent into a solution separated from it by a semipermeable membrane



Thermodynamics of Osmosis



$$\mu_A^*(p) = \mu_A(x_A, p + \Pi)$$

$$\mu_A^*(p) = \mu_A^*(p + \Pi) + RT \ln x_A$$

Thus,

$$\mu_A^*(p + \Pi) = \mu_A^*(p) - RT \ln x_A$$

Pressure Effect

In general,

$$G_m(p_2) = G_m(p_1) + \int_{p_1}^{p_2} V_m dp$$

$$\mu_A^*(p + \Pi) = \mu_A^*(p) + \int_p^{p+\Pi} V_m dp'$$

Assuming that V_m is constant,

$$\mu_A^*(p + \Pi) = \mu_A^*(p) + V_m \int_p^{p+\Pi} 1 dp' = \mu_A^*(p) + V_m \Pi$$

Combining the Two

$$\mu_A^*(p + \Pi) = \mu_A^*(p) - RT \ln x_A$$

$$\mu_A^*(p + \Pi) = \mu_A^*(p) + V_m \Pi$$

Comparing the two equations,

$$-RT \ln x_A = V_m \Pi$$

$$-RT \ln(1 - x_B) = V_m \Pi$$

Since $\ln(1 - x_B) \approx -x_B$ for $x_B \ll 1$,

$$RT x_B = V_m \Pi$$

Osmotic Pressure

$$RTx_B = V_m\Pi$$

For a dilute solution,

$$x_B = n_B/(n_A + n_B) \approx n_B/n_A$$

Thus,

$$RTn_B = n_A V_m \Pi = V \Pi$$

and

$$\Pi = [B]RT \text{ (van't Hoff equation)}$$

Real Solutions

For an ideal solution (now solute = J),

$$\Pi = [J]RT$$

The osmotic virial expansion is

$$\Pi = [J]RT(1 + B[J] + \dots)$$

where B is called the **osmotic virial coefficient**.

Osmometry

$$\Pi = [J]RT(1 + B[J] + \dots)$$

Taking the first correction term only,

$$\Pi \approx [J]RT(1 + B[J])$$

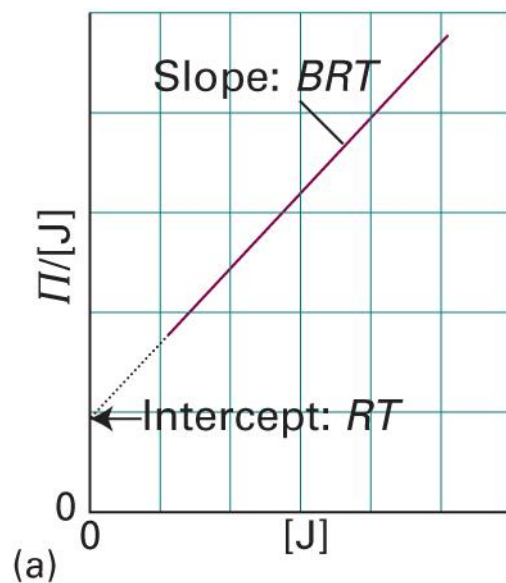
$$\frac{\Pi}{[J]} \approx RT + BRT[J]$$

The mass concentration is $c_{\text{mass},J} = [J]M$;

$$\frac{\Pi}{c_{\text{mass},J}} \approx \frac{RT}{M} + \frac{BRT}{M^2} c_{\text{mass},J}$$

Osmometry

$$\Pi/[J] \approx RT + BRT[J]$$



$$\frac{\Pi}{c_{\text{mass},J}} \approx \frac{RT}{M} + \frac{BRT}{M^2} c_{\text{mass},J}$$

